Practical exercise 4

1. Introduction to bootstrap.

   Consider the sample data below, giving the results of 13 tests:

   ```
   23  22  20  23  19  19  16  17  25  20  21  22  20
   ```

   Input these into a vector called 'results' in `R` (for example by using `results=scan()`).

   Produce a summary and a graph of these data.

   Obtain a sample with replacement of the same size as the vector 'results'. Use the code:

   ```
   results1 = sample(results, size=13, replace=TRUE)
   ```

   Compare the two sets of results graphically and numerically.

   We will now focus on the mean parameter and generate 1000 sets of results. Use:

   ```
   res = numeric(1000)
   for(i in 1:1000) res[i] = mean(sample(results, size=13, replace=TRUE))
   ```

   Then look at your 1000 means using

   ```
   res
   ```

   Obtain a histogram of your results.

   A 95% confidence interval for the population mean in a bootstrap problem can be obtained by using:

   ```
   quantile(res,c(0.025,0.975))
   ```

   Compare this confidence interval with the one you can obtain using the `t` distribution. The command for that is:

   ```
   t.test(results)
   ```

2. Bootstrap confidence interval for a proportion.

   The data below represents survival (1) or death (0) in 5 years for a random sample of 52 patients under age 40 with lung cancer:

```
0 0 0 0 1 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 0 0 1
```

   Notice that the proportion that survive five years is the same as the mean of the 0's and 1's. We can therefore find a bootstrap confidence interval for the population proportion that survive five years using the same methods as in question 1.

   (a) Produce the bootstrap sampling distribution. Does it look symmetric? Obtain a 95% confidence interval for the population proportion that survive five years.

   (b) Is the sample size large enough to approximate the binomial distribution with a Normal? Explain. Calculate the 95% confidence interval based on a normal approximation to the binomial distribution. How do the two intervals compare?

   Reminder: for proportions when n is large, $\frac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$ approximately.

3. Hypothesis tests using bootstrap.

   Consider again the air conditioning failures data (given below) discussed in lectures.

```
413 14 58 37 100 65 9 169 447 184 36 201
118 34 31 18 18 67 57 62 7 22 34
```

   Let σ^2 denote the variance of the corresponding population. We want to test the hypotheses:

$$H_0 : \sigma^2 = 9000 \;\; v \;\; H_1 : \sigma^2 > 9000.$$

   (a) Use parametric bootstrap methodology (as presented in lectures) to perform this test.

   (b) Compare the p-value and conclusion from part (a) with those obtained by performing the test using the sampling distribution

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

   Comment on the validity of the two tests.